PROJECT 1 — CUSTOMER SEGMENTATION & RFM ANALYSIS

-- 1.1 Total customer and order count

```
SELECT
  COUNT(DISTINCT customerid)   AS uniquecustomers,
  COUNT(orderid)               AS totalorders,
  ROUND(
    COUNT(orderid) * 1.0 /
    COUNT(DISTINCT customerid), 2
  )                            AS avgorderspercustomer,
  MIN(date)                    AS firstorderdate,
  MAX(date)                    AS latestorderdate
FROM ecommerce;
--
```

	123 uniquecustomers	123 totalorders	123 avgorderspercustomer	firstorderdate	latestorderdate
1	1,189	1,200	1.01	2023-01-01	2025-06-30

-- 1.2 Repeat vs one-time buyer split

```
WITH customerorders AS (
  SELECT
    customerid,
    COUNT(orderid) AS totalorders
  FROM ecommerce
  GROUP BY customerid
)
SELECT
  CASE WHEN totalorders = 1 THEN 'One-Time Buyer'
     ELSE 'Repeat Buyer'
```

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```
  END                          AS customertype,
  COUNT(*)                     AS customercount,
  ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (), 1) AS pctofcustomers
FROM customerorders
GROUP BY customertype
ORDER BY customercount DESC;
--
```

	customertype	customercount	pctofcustomers
1	One-Time Buyer	1,178	99.1
2	Repeat Buyer	11	0.9

-- 1.3 Customer revenue distribution

```
WITH customerrevenue AS (
  SELECT
    customerid,
    ROUND(SUM(totalprice), 2) AS lifetimevalue
  FROM ecommerce
  GROUP BY customerid
)
SELECT
  CASE
    WHEN lifetimevalue >= 3000 THEN 'High Value   (>= $3,000)'
    WHEN lifetimevalue >= 1500 THEN 'Mid Value    ($1,500–$2,999)'
    WHEN lifetimevalue >= 500  THEN 'Low Value    ($500–$1,499)'
    ELSE                 'Minimal Value (< $500)'
  END                          AS valuetier,
  COUNT(*)                     AS customers,
  ROUND(SUM(lifetimevalue), 2)          AS totalrevenue,
```

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```
  ROUND(SUM(lifetimevalue) * 100.0 /
    SUM(SUM(lifetimevalue)) OVER (), 1)      AS pctofrevenue
FROM customerrevenue
GROUP BY valuetier
ORDER BY totalrevenue DESC;
--
```

	valuetier	customers	totalrevenue	pctofrevenue
1	Mid Value ($1,500–$2,999)	289	602,379.98	47.6
2	Low Value ($500–$1,499)	493	451,172.43	35.7
3	High Value (>= $3,000)	34	112,184.68	8.9
4	Minimal Value (< $500)	373	99,024.87	7.8

```
/* ==============================================================================
  SECTION 2 — RFM METRIC CALCULATION
  ==============================================================================
*/
```

-- 2.1 Calculate raw RFM metrics per customer

```
CREATE OR REPLACE VIEW rfmbase AS
SELECT
  customerid,
  MAX(date)                    AS lastorderdate,
  COUNT(orderid)               AS frequency,
  ROUND(SUM(totalprice), 2)    AS monetary,
  DATE '2025-06-30' - MAX(date)   AS recencydays
FROM ecommerce
GROUP BY customerid;
```

-- View the base RFM metrics

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```
SELECT *
FROM rfmbase
ORDER BY monetary DESC
LIMIT 20;
```

	customerid	lastorderdate	frequency	monetary	recencydays
1	C38840	2024-06-08	2	5,723.23	387
2	C57276	2023-08-17	1	3,456.4	683
3	C67260	2023-05-02	1	3,390.8	790
4	C13877	2023-11-26	1	3,384.9	582
5	C18404	2023-02-28	1	3,370.2	853
6	C16775	2023-03-27	1	3,353.75	826
7	C65986	2024-07-01	1	3,352.4	364
8	C47778	2023-10-30	1	3,334	609
9	C59183	2023-02-28	1	3,322.55	853
10	C25276	2023-05-26	1	3,313.9	766

-- 2.2 RFM distribution summary

```
SELECT
  ROUND(MIN(recencydays), 0)  AS minrecency,
  ROUND(AVG(recencydays), 0)  AS avgrecency,
  ROUND(MAX(recencydays), 0)  AS maxrecency,
  ROUND(MIN(frequency), 0)  AS minfrequency,
  ROUND(AVG(frequency), 2)  AS avgfrequency,
  ROUND(MAX(frequency), 0)  AS maxfrequency,
  ROUND(MIN(monetary), 2)  AS minmonetary,
  ROUND(AVG(monetary), 2)  AS avgmonetary,
  ROUND(MAX(monetary), 2)  AS maxmonetary
FROM rfmbase;
```

--

	minrecency	avgrecency	maxrecency	minfrequency	avgfrequency	maxfrequency	minmonetary	avgmonetary	maxmonetary
1	0	462	911	1	1.01	2	11.39	1,063.72	5,723.23

/* ==========

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SECTION 3 — RFM SCORING

==========
*/

-- 3.1 Score each customer on R, F, M dimensions

```
CREATE OR REPLACE VIEW rfmscores AS
SELECT
  customerid,
  lastorderdate,
  recencydays,
  frequency,
  monetary,
  5 - NTILE(4) OVER (ORDER BY recencydays ASC) AS rscore,
  NTILE(4) OVER (ORDER BY frequency ASC)      AS fscore,
  NTILE(4) OVER (ORDER BY monetary ASC)       AS mscore
FROM rfmbase;
```

-- 3.2 View RFM scores with combined total

```
SELECT
  customerid,
  recencydays,
  frequency,
  ROUND(monetary, 2)        AS monetary,
  rscore,
  fscore,
  mscore,
  (rscore + fscore + mscore)   AS rfmtotalscore
FROM rfmscores
```

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ORDER BY rfmtotalscore **DESC**

LIMIT 20;

	customerid	recencydays	frequency	monetary	rscore	fscore	mscore	rfmtotalscore
1	C43595	48	1	2,807.4	4	4	4	12
2	C71698	44	1	2,446.85	4	4	4	12
3	C97681	23	1	2,166.3	4	4	4	12
4	C78725	29	1	1,880.36	4	4	4	12
5	C50236	36	1	2,463.32	4	4	4	12
6	C91591	43	1	2,609.92	4	4	4	12
7	C46886	22	1	1,849.17	4	4	4	12
8	C88029	16	1	3,277.75	4	4	4	12
9	C63827	25	1	1,974.92	4	4	4	12
10	C30686	17	1	1,614.32	4	4	4	12

-- 3.3 Score distribution

SELECT

(rscore + fscore + mscore) **AS** rfmtotalscore,

COUNT(*) **AS** customercount

FROM rfmscores

GROUP BY rfmtotalscore

ORDER BY rfmtotalscore **DESC**;

	rfmtotalscore	customercount
1	12	62
2	11	76
3	10	158
4	9	140
5	8	162
6	7	159
7	6	140
8	5	145
9	4	81
10	3	66

--

/* ==========

SECTION 4 — CUSTOMER SEGMENTATION

```
  ==========================================================================
*/
```

-- 4.1 Assign segments based on combined RFM score

```
CREATE OR REPLACE VIEW rfmsegments AS
SELECT
  customerid,
  lastorderdate,
  recencydays,
  frequency,
  monetary,
  rscore,
  fscore,
  mscore,
  (rscore + fscore + mscore) AS rfmtotalscore,
  CASE
    WHEN (rscore + fscore + mscore) >= 10 THEN 'Champion'
    WHEN (rscore + fscore + mscore) >= 8  THEN 'Loyal'
    WHEN (rscore + fscore + mscore) >= 6  THEN 'Potential Loyal'
    WHEN (rscore + fscore + mscore) >= 4  THEN 'At Risk'
    ELSE                                  'Lost'
  END AS segment,
  CASE
    WHEN (rscore + fscore + mscore) >= 10 THEN 'Reward heavily. They are your best customers.'
    WHEN (rscore + fscore + mscore) >= 8  THEN 'Nurture with loyalty program and early access.'
    WHEN (rscore + fscore + mscore) >= 6  THEN 'Engage with personalised offers to convert.'
    WHEN (rscore + fscore + mscore) >= 4  THEN 'Win-back campaign before they go cold.'
    ELSE                                  'Last-chance offer. Accept churn if no response.'
  END AS recommendedaction
```

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```
FROM rfmscores;
```

-- 4.2 Full segment overview

```
SELECT
  segment,
  COUNT(customerid) AS customercount,
  ROUND(COUNT(customerid) * 100.0 /
    SUM(COUNT(customerid)) OVER (), 1) AS pctofcustomers,
  ROUND(AVG(recencydays), 0)        AS avgrecencydays,
  ROUND(AVG(frequency), 2)          AS avgorders,
  ROUND(AVG(monetary), 2)           AS avglifetimevalue,
  ROUND(SUM(monetary), 2)           AS totalsegmentrevenue
FROM rfmsegments
GROUP BY segment
ORDER BY avglifetimevalue DESC;
--
```

	segment	customercount	pctofcustomers	avgrecencydays	avgorders	avglifetimevalue	totalsegmentrevenue
1	Champion	296	24.9	285	1.03	1,928.49	570,832.65
2	Loyal	302	25.4	462	1.01	1,282.21	387,227.61
3	Potential Loyal	299	25.1	479	1	728.3	217,760.27
4	At Risk	226	19	573	1	336.95	76,151.1
5	Lost	66	5.6	801	1	193.79	12,790.33

```
/* ==========================================================================
  SECTION 5 — SEGMENT REVENUE & BUSINESS VALUE ANALYSIS
  ==========================================================================
*/
```

-- 5.1 Revenue concentration by segment

```
SELECT
  segment,
  COUNT(customerid)                    AS customers,
```

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```
  ROUND(SUM(monetary), 2)                    AS totalrevenue,
  ROUND(SUM(monetary) * 100.0 / SUM(SUM(monetary)) OVER (), 1) AS pctoftotalrevenue,
  ROUND(AVG(monetary), 2)                    AS avgcustomervalue,
  ROUND(MIN(monetary), 2)                    AS minvalue,
  ROUND(MAX(monetary), 2)                    AS maxvalue
FROM rfmsegments
GROUP BY segment
ORDER BY totalrevenue DESC;
--
```

	segment	customers	totalrevenue	pctoftotalrevenue	avgcustomervalue	minvalue	maxvalue
1	Champion	296	570,832.65	45.1	1,928.49	647.77	5,723.23
2	Loyal	302	387,227.61	30.6	1,282.21	418.35	3,352.4
3	Potential Loyal	299	217,760.27	17.2	728.3	11.39	3,456.4
4	At Risk	226	76,151.1	6	336.95	14.06	815.58
5	Lost	66	12,790.33	1	193.79	17.24	405.28

-- 5.2 Segment value-to-size efficiency ratio

```
SELECT
  segment,
  COUNT(customerid)                    AS customers,
  ROUND(SUM(monetary), 2)                    AS totalrevenue,
  ROUND(SUM(monetary) / COUNT(customerid), 2)       AS revenuepercustomer,
  ROUND(
    (SUM(monetary) / COUNT(customerid)) /
    AVG(SUM(monetary) / COUNT(customerid)) OVER (),
    2
  )                                    AS efficiencyvsavg
FROM rfmsegments
GROUP BY segment
ORDER BY revenuepercustomer DESC;
```

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	segment	customers	totalrevenue	revenuepercustomer	efficiencyvsavg
1	Champion	296	570,832.65	1,928.49	2.16
2	Loyal	302	387,227.61	1,282.21	1.43
3	Potential Loyal	299	217,760.27	728.3	0.81
4	At Risk	226	76,151.1	336.95	0.38
5	Lost	66	12,790.33	193.79	0.22

-- 5.3 Estimated revenue at risk

```
SELECT
  'At Risk + Lost Customers'    AS scenario,
  COUNT(customerid)             AS customersatrisk,
  ROUND(SUM(monetary), 2)       AS revenueatrisk,
  ROUND(SUM(monetary) * 0.20, 2)  AS recovery20pcttarget,
  ROUND(SUM(monetary) * 0.35, 2)  AS recovery35pcttarget
FROM rfmsegments
WHERE segment IN ('At Risk', 'Lost');
```

--

	scenario	customersatrisk	revenueatrisk	recovery20pcttarget	recovery35pcttarget
1	At Risk + Lost Customers	292	88,941.43	17,788.29	31,129.5

```
/* ==========================================================================
  SECTION 6 — COUPON EFFECTIVENESS BY SEGMENT
  ==========================================================================
*/
```

-- 6.1 Coupon usage distribution across segments

```
SELECT
  s.segment,
  COALESCE(o.couponcode, 'NO COUPON') AS couponcode,
  COUNT(o.orderid)              AS orderswithcoupon,
```

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```
  ROUND(SUM(o.totalprice), 2)      AS revenuewithcoupon,
  ROUND(AVG(o.totalprice), 2)      AS avgordervalue
FROM rfmsegments s
JOIN ecommerce o ON s.customerid = o.customerid
GROUP BY s.segment, couponcode
ORDER BY s.segment, orderswithcoupon DESC;
--
```

	segment	couponcode	orderswithcoupon	revenuewithcoupon	avgordervalue
1	At Risk		57	19,208.28	336.99
2	At Risk	SAVE10	57	17,032.81	298.82
3	At Risk	FREESHIP	56	20,574.93	367.41
4	At Risk	WINTER15	56	19,335.08	345.27
5	Champion	FREESHIP	92	178,035.4	1,935.17
6	Champion	SAVE10	74	131,517.17	1,777.26
7	Champion	WINTER15	71	133,455.03	1,879.65
8	Champion		67	127,825.05	1,907.84
9	Lost	WINTER15	19	3,550.16	186.85
10	Lost	SAVE10	18	4,211.95	234

-- 6.2 Which segments use coupons most

```
SELECT
  s.segment,
  COUNT(o.orderid) AS totalorders,
  SUM(CASE WHEN o.couponcode IS NOT NULL THEN 1 ELSE 0 END) AS couponorders,
  ROUND(
    SUM(CASE WHEN o.couponcode IS NOT NULL THEN 1 ELSE 0 END)
    * 100.0 / COUNT(o.orderid), 1
  ) AS couponusagepct,
  ROUND(AVG(CASE WHEN o.couponcode IS NOT NULL
        THEN o.totalprice END), 2) AS avgcouponordervalue,
  ROUND(AVG(CASE WHEN o.couponcode IS NULL
        THEN o.totalprice END), 2) AS avgnoncouponordervalue
FROM rfmsegments s
```

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```
JOIN ecommerce o ON s.customerid = o.customerid
GROUP BY s.segment
ORDER BY couponusagepct DESC;
--
```

	segment	totalorders	couponorders	couponusagepct	avgcouponordervalue	avgnoncouponordervalue
1	Champion	304	304	100	1,877.74	[NULL]
2	Loyal	304	304	100	1,273.78	[NULL]
3	Lost	66	66	100	193.79	[NULL]
4	At Risk	226	226	100	336.95	[NULL]
5	Potential Loyal	300	300	100	725.87	[NULL]

-- 6.3 FREESHIP coupon analysis

```
SELECT
  s.segment,
  COUNT(CASE WHEN o.couponcode = 'FREESHIP' THEN 1 END) AS freeshiporders,
  COUNT(CASE WHEN o.couponcode = 'FREESHIP'
        AND o.orderstatus = 'Delivered' THEN 1 END) AS freeshipdelivered,
  COUNT(CASE WHEN o.couponcode = 'FREESHIP'
        AND o.orderstatus = 'Returned' THEN 1 END) AS freeshipreturned,
  ROUND(
    COUNT(CASE WHEN o.couponcode = 'FREESHIP'
          AND o.orderstatus = 'Returned' THEN 1 END)
    * 100.0 / NULLIF(
      COUNT(CASE WHEN o.couponcode = 'FREESHIP' THEN 1 END), 0
    ), 1
  ) AS freeshipreturnratepct
FROM rfmsegments s
JOIN ecommerce o ON s.customerid = o.customerid
GROUP BY s.segment
ORDER BY freeshipreturnratepct DESC NULLS LAST;
```

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	segment	freeshiporders	freeshipdelivered	freeshipreturned	freeshipreturnratepct
1	At Risk	56	7	12	21.4
2	Potential Loyal	88	21	18	20.5
3	Champion	92	17	18	19.6
4	Loyal	64	13	11	17.2
5	Lost	13	3	2	15.4

/* ==========

SECTION 7 — ACQUISITION CHANNEL BY SEGMENT

==========
*/

-- 7.1 Segment distribution by referral source

```
SELECT
  o.referralsource,
  s.segment,
  COUNT(DISTINCT s.customerid) AS customers
FROM rfmsegments s
JOIN ecommerce o ON s.customerid = o.customerid
GROUP BY o.referralsource, s.segment
ORDER BY o.referralsource, customers DESC;
```

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	referralsource	segment	customers
1	Email	Potential Loyal	64
2	Email	Champion	61
3	Email	Loyal	61
4	Email	At Risk	50
5	Email	Lost	14
6	Facebook	Loyal	70
7	Facebook	Champion	58
8	Facebook	Potential Loyal	54
9	Facebook	At Risk	35
10	Facebook	Lost	11

--

-- 7.2 Channel quality score

```
SELECT
  o.referralsource,
  COUNT(DISTINCT s.customerid) AS totalcustomers,
  COUNT(DISTINCT CASE WHEN s.segment IN ('Champion','Loyal')
            THEN s.customerid END) AS highvaluecustomers,
  ROUND(
    COUNT(DISTINCT CASE WHEN s.segment IN ('Champion','Loyal')
              THEN s.customerid END)
    * 100.0 / COUNT(DISTINCT s.customerid), 1
  ) AS highvaluepct,
  COUNT(DISTINCT CASE WHEN s.segment IN ('At Risk','Lost')
            THEN s.customerid END) AS lowqualitycustomers,
  ROUND(
    COUNT(DISTINCT CASE WHEN s.segment IN ('At Risk','Lost')
              THEN s.customerid END)
```

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```
    * 100.0 / COUNT(DISTINCT s.customerid), 1
  ) AS lowqualitypct,
  ROUND(AVG(s.monetary), 2) AS avgcustomerltv
FROM rfmsegments s
JOIN ecommerce o ON s.customerid = o.customerid
GROUP BY o.referralsource
ORDER BY highvaluepct DESC;
--
```

	referralsource	totalcustomers	highvaluecustomers	highvaluepct	lowqualitycustomers	lowqualitypct	avgcustomerltv
1	Facebook	228	128	56.1	46	20.2	1,120.37
2	Referral	221	112	50.7	58	26.2	1,045.24
3	Google	241	121	50.2	59	24.5	1,051.08
4	Email	250	122	48.8	64	25.6	1,052.55
5	Instagram	257	122	47.5	65	25.3	1,082.48

-- 7.3 Best channel + segment combination

```
SELECT
  o.referralsource,
  s.segment,
  COUNT(DISTINCT s.customerid) AS customers,
  ROUND(AVG(s.monetary), 2)   AS avglifetimevalue,
  ROUND(SUM(s.monetary), 2)   AS totalrevenue
FROM rfmsegments s
JOIN ecommerce o ON s.customerid = o.customerid
GROUP BY o.referralsource, s.segment
ORDER BY avglifetimevalue DESC
LIMIT 15;
```

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	referralsource	segment	customers	avglifetimevalue	totalrevenue
1	Email	Champion	61	1,998.78	121,925.48
2	Google	Champion	65	1,943.78	126,345.8
3	Instagram	Champion	64	1,935.73	127,758.26
4	Facebook	Champion	58	1,934.42	112,196.21
5	Referral	Champion	53	1,849.7	99,883.97
6	Instagram	Loyal	58	1,339.16	77,671.29
7	Facebook	Loyal	70	1,317.32	92,212.32
8	Referral	Loyal	59	1,279.16	75,470.6
9	Email	Loyal	61	1,244.21	75,896.91
10	Google	Loyal	56	1,215.04	68,042.2

```
/* ==========================================================================
  SECTION 8 — REPEAT PURCHASE & RETENTION ANALYSIS
  ==========================================================================
*/
```

-- 8.1 Repeat purchase rate overall

```
WITH customerordercount AS (
  SELECT
    customerid,
    COUNT(orderid) AS ordercount
  FROM ecommerce
  GROUP BY customerid
)
SELECT
  SUM(CASE WHEN ordercount >= 2 THEN 1 ELSE 0 END) AS repeatbuyers,
  SUM(CASE WHEN ordercount = 1 THEN 1 ELSE 0 END)  AS onetimebuyers,
  COUNT(*)                                         AS totalcustomers,
  ROUND(
    SUM(CASE WHEN ordercount >= 2 THEN 1 ELSE 0 END)
    * 100.0 / COUNT(*), 2
```

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```
  ) AS repeatratepct
FROM customerordercount;
--
```

	123 repeatbuyers	123 onetimebuyers	123 totalcustomers	123 repeatratepct
1	11	1,178	1,189	0.93

-- 8.2 Days between first and second order

```
WITH ordersequence AS (
  SELECT
    customerid,
    date,
    ROW_NUMBER() OVER (PARTITION BY customerid ORDER BY date) AS orderrank
  FROM ecommerce
)
SELECT
  firstorder.customerid,
  firstorder.date AS firstorderdate,
  secondorder.date AS secondorderdate,
  secondorder.date - firstorder.date AS daystorepurchase
FROM ordersequence firstorder
JOIN ordersequence secondorder
  ON firstorder.customerid = secondorder.customerid
  AND firstorder.orderrank = 1
  AND secondorder.orderrank = 2
ORDER BY daystorepurchase ASC;
```

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	customerid	firstorderdate	secondorderdate	daystorepurchase
1	C94569	2024-03-23	2024-06-21	90
2	C56969	2023-06-29	2024-01-09	194
3	C46651	2023-08-01	2024-03-20	232
4	C97593	2023-05-03	2024-02-22	295
5	C35852	2024-03-05	2025-03-02	362
6	C38840	2023-06-07	2024-06-08	367
7	C21191	2024-04-11	2025-05-16	400
8	C91155	2023-07-17	2024-09-08	419
9	C98474	2023-03-27	2024-06-03	434
10	C70659	2023-08-20	2025-03-01	559

-- 8.3 Recency buckets

```
SELECT
  CASE
    WHEN recencydays <= 90  THEN '0–90 days   (Active)'
    WHEN recencydays <= 180 THEN '91–180 days (Warm)'
    WHEN recencydays <= 365 THEN '181–365 days (Cooling)'
    WHEN recencydays <= 730 THEN '366–730 days (Cold)'
    ELSE                         '730+ days   (Dormant)'
  END AS recencybucket,
  COUNT(*) AS customers,
  ROUND(AVG(monetary), 2) AS avglifetimevalue,
  ROUND(SUM(monetary), 2) AS totalrevenue,
  ROUND(COUNT(*) * 100.0 /
    SUM(COUNT(*)) OVER (), 1) AS pctofcustomers
FROM rfmsegments
GROUP BY recencybucket
ORDER BY MIN(recencydays);
```

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	recencybucket	customers	avglifetimevalue	totalrevenue	pctofcustomers
1	0–90 days (Active)	118	1,089.72	128,587.2	9.9
2	91–180 days (Warm)	113	920.98	104,070.27	9.5
3	181–365 days (Cooling)	223	1,014.91	226,324.12	18.8
4	366–730 days (Cold)	488	1,078.31	526,213.95	41
5	730+ days (Dormant)	247	1,131.85	279,566.42	20.8

/* ==========

SECTION 9 — AT-RISK CUSTOMER IDENTIFICATION

==========
*/

-- 9.1 Priority win-back list

```
SELECT
  customerid,
  lastorderdate,
  recencydays,
  frequency,
  ROUND(monetary, 2) AS lifetimevalue,
  rscore,
  fscore,
  mscore,
  rfmtotalscore,
  segment,
  recommendedaction
FROM rfmsegments
WHERE segment = 'At Risk'
ORDER BY monetary DESC
LIMIT 25;
```

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	customerid	lastorderdate	recencydays	frequency	lifetimevalue	rscore	fscore	mscore	rfmtotalscore	segment
1	C12245	2023-05-17	775	1	815.58	1	2	2	5	At Risk
2	C85322	2023-05-01	791	1	813.06	1	2	2	5	At Risk
3	C86773	2023-06-09	752	1	789.44	1	2	2	5	At Risk
4	C29314	2023-06-08	753	1	788.6	1	2	2	5	At Risk
5	C59685	2023-01-26	886	1	788.32	1	2	2	5	At Risk
6	C44366	2023-01-04	908	1	776.44	1	2	2	5	At Risk
7	C88304	2023-05-27	765	1	768.28	1	2	2	5	At Risk
8	C41460	2023-05-12	780	1	747.75	1	2	2	5	At Risk
9	C79506	2023-07-18	713	1	746.34	1	2	2	5	At Risk
10	C46504	2023-02-21	860	1	745.84	1	2	2	5	At Risk

	recencydays	frequency	lifetimevalue	rscore	fscore	mscore	rfmtotalscore	segment	recommendedaction
05-17	775	1	815.58	1	2	2	5	At Risk	Win-back campaign before they go cold.
05-01	791	1	813.06	1	2	2	5	At Risk	Win-back campaign before they go cold.
06-09	752	1	789.44	1	2	2	5	At Risk	Win-back campaign before they go cold.
06-08	753	1	788.6	1	2	2	5	At Risk	Win-back campaign before they go cold.
01-26	886	1	788.32	1	2	2	5	At Risk	Win-back campaign before they go cold.
01-04	908	1	776.44	1	2	2	5	At Risk	Win-back campaign before they go cold.
05-27	765	1	768.28	1	2	2	5	At Risk	Win-back campaign before they go cold.
05-12	780	1	747.75	1	2	2	5	At Risk	Win-back campaign before they go cold.
07-18	713	1	746.34	1	2	2	5	At Risk	Win-back campaign before they go cold.
02-21	860	1	745.84	1	2	2	5	At Risk	Win-back campaign before they go cold.

-- 9.2 Full prioritised retention action list

```
SELECT
  segment,
  customerid,
  ROUND(monetary, 2) AS lifetimevalue,
  recencydays,
  rfmtotalscore,
  recommendedaction
FROM rfmsegments
WHERE segment IN ('Champion', 'Loyal', 'At Risk')
ORDER BY
  CASE segment
    WHEN 'Champion' THEN 1
    WHEN 'Loyal' THEN 2
    WHEN 'At Risk' THEN 3
  END,
  monetary DESC;
```

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--

	segment	customerid	lifetimevalue	recencydays	rfmtotalscore	recommendedaction
1	Champion	C38840	5,723.23	387	11	Reward heavily. They are your best customers.
2	Champion	C88029	3,277.75	16	12	Reward heavily. They are your best customers.
3	Champion	C27202	3,267.35	315	11	Reward heavily. They are your best customers.
4	Champion	C35987	3,267.3	665	10	Reward heavily. They are your best customers.
5	Champion	C36725	3,253.6	246	11	Reward heavily. They are your best customers.
6	Champion	C87281	3,243.25	518	10	Reward heavily. They are your best customers.
7	Champion	C84331	3,223.2	499	10	Reward heavily. They are your best customers.
8	Champion	C88205	3,215.15	376	11	Reward heavily. They are your best customers.
9	Champion	C93473	3,196.85	493	10	Reward heavily. They are your best customers.
10	Champion	C52879	3,194.6	55	12	Reward heavily. They are your best customers.

-- 9.3 Segment-level estimated win-back revenue potential

```
SELECT
  segment,
  COUNT(customerid) AS customers,
  ROUND(SUM(monetary), 2) AS totalltv,
  ROUND(SUM(monetary) * 0.15, 2) AS potentialif15pctretained,
  ROUND(SUM(monetary) * 0.25, 2) AS potentialif25pctretained,
  ROUND(SUM(monetary) * 0.40, 2) AS potentialif40pctretained
FROM rfmsegments
WHERE segment IN ('At Risk', 'Potential Loyal')
GROUP BY segment
ORDER BY totalltv DESC;
```

--

	segment	customers	totalltv	potentialif15pctretained	potentialif25pctretained	potentialif40pctretained
1	Potential Loyal	299	217,760.27	32,664.04	54,440.07	87,104.11
2	At Risk	226	76,151.1	11,422.67	19,037.78	30,460.44

MD IBRAHIM HASAN JALANI